

Graph-conditioned state-space models for genetic perturbation response prediction: a controlled evaluation against simple baselines

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Abstract

Motivation: Predicting the transcriptional response to unmeasured genetic perturbations—in particular to combinations of perturbations—is a central goal of single-cell perturbation modelling, yet recent benchmarks show that deep models rarely beat simple additive or linear baselines. State-space models (SSMs) scale linearly with the number of genes, but genes have no natural order and an SSM has no notion of which genes are related. We asked whether conditioning an SSM on a gene–gene co-expression graph closes this gap.

Results: We introduce GCP-Mamba, a bidirectional selective SSM over all 5,045 measured genes in which the diffusion of the perturbation over a control-cell co-expression graph modulates the discretization step of every gene token. On the five standard GEARS splits of the Norman et al. CRISPRa screen, with three initialisations per split, GCP-Mamba clearly improved on predicting no change or the mean response for double perturbations, but it was less accurate than the sum of single-perturbation effects, and it was not more accurate than the mean response, a linear model or its own graph-free and graph-permuted ablations for unseen single perturbations. Conditioning the step size on the graph had no measurable effect. When the network predicted only the deviation from the additive sum, it matched but did not improve on the additive prediction: no model explained any genetic-interaction variance beyond additivity. GEARS, retrained on one split, predicted unseen targets more accurately than GCP-Mamba but was also less accurate than the additive sum overall. On the Adamson et al. CRISPRi screen all methods were within noise of the mean response. We also document two evaluation pitfalls: a perturbation encoding restricted to highly variable genes that silently gave 80% of training perturbations an all-zero input in an earlier version of this work, and a genetic-interaction correlation metric that ranked models in the reverse order of the variance they explained.

Availability and implementation: Code, splits and all per-condition results are available at <https://github.com/The-MathKing/gcpmamba> (MIT license).

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Supplementary information: Supplementary data are available at *Bioinformatics* online.

1 Introduction

Perturb-seq screens measure the transcriptome-wide response of single cells to CRISPR-mediated gene perturbations (Dixit et al., 2016; Norman et al., 2019; Replogle et al., 2022). Because the number of possible combinations grows quadratically with the number of targets, only a small fraction of double perturbations can ever be measured: the Norman et al. K562 CRISPRa screen profiled 131 of the $\binom{105}{2} = 5,460$ pairs of its 105 targets (Norman et al., 2019). Models that predict the response to unmeasured perturbations—and in particular the non-additive, genetic-interaction (GI) component of double perturbations—are therefore a central goal of the field.

Knowledge-graph neural networks such as GEARS (Roohani et al., 2024) and TxPert (Wenkel et al., 2026), and transformer-based single-cell foundation models (Cui et al., 2024; Theodoris et al., 2023), have been proposed for this task. However, recent benchmarks found that such models often do not outperform deliberately simple baselines: the mean training response, the sum of single-perturbation effects, or a low-rank linear model (Ahlmann-Eltze et al., 2025; Csentesi et al., 2025). Two lessons from this literature shape our study: every new architecture must be compared against these baselines on identical splits, and the contribution of any biological prior must be isolated by controls that remove it while keeping everything else fixed.

State-space models (SSMs) such as Mamba (Gu and Dao, 2023) process sequences in time and memory linear in their length, whereas self-attention is quadratic (Vaswani et al., 2017). Treating the transcriptome as a sequence of gene tokens, an SSM can process all measured genes at

once, but genes have no natural order and the recurrence has no notion of which genes are functionally related. We asked whether both limitations can be addressed by conditioning the SSM’s discretization step on a gene–gene graph, and—more importantly—whether doing so yields better predictions than simple baselines.

Our contributions are: (i) GCP-Mamba, a bidirectional selective SSM over all measured genes in which the diffusion of the perturbation over a co-expression graph modulates the step size of every gene token; (ii) a controlled benchmark on the five standard GEARS splits of the Norman data and on the Adamson et al. CRISPRi screen, against five simple baselines, GEARS retrained on one of the splits, and eight ablations and variants that remove, permute or relocate the graph, destroy the gene ordering, remove the recurrence or add an additive prior; and (iii) two evaluation pitfalls, one of which invalidated the conclusions of an earlier version of this work.

2 Methods

2.1 Data and splits

We used the Norman et al. K562 CRISPRa Perturb-seq data (Norman et al., 2019) as distributed with GEARS: 91,205 cells, 5,045 genes, 7,353 control cells, single perturbations of 105 genes and 131 double perturbations, log-normalised. For every condition p we computed the pseudobulk response $\delta_p = \bar{x}_p - \bar{x}_{\text{ctrl}} \in \mathbb{R}^{5045}$. Singles are recorded as two conditions (A+ctrl, ctrl+A) that use different guides; wherever a single response is needed we average them. We used the GEARS *simulation* splits with seeds 1–5, generated with the official `cell-gears` code (Roohani et al., 2024). Each holds out about a quarter of the targets; test perturbations are grouped into unseen singles and doubles with zero, one or two targets seen during training (“0/2”, “1/2”, “2/2 seen”; GEARS counts a target as seen if it appears in the training or validation set). Across the five splits the test sets contain 185 unseen-single, 32 “0/2”, 254 “1/2” and 90 “2/2” perturbations. Validation perturbations were used only for early stopping and for choosing the ridge penalty of the linear baselines. As a second dataset we used the Adamson et al. K562 CRISPRi screen (Adamson et al., 2016) (86 single perturbations, 5,060 genes; GEARS simulation splits, seeds 1–5).

The co-expression graph, gene embeddings and gene order below were computed from unperturbed control cells only; they are identical in every split and contain no information about any perturbation response.

2.2 Co-expression graph and perturbation signal

We z-scored control-cell expression, computed absolute Pearson correlations between all gene pairs, kept the $k = 20$ strongest neighbours of every gene and symmetrised the result, giving a sparse graph $\mathbf{W}$ with 111,078 edges. For a perturbation set P with indicator $\mathbf{1}_P \in \{0, 1\}^G$, the graph signal is a truncated diffusion

$$c(P) = \log\left(1 + s \sum_{k=1}^3 \beta^{k-1} \widetilde{\mathbf{W}}^k \mathbf{1}_P\right), \quad \widetilde{\mathbf{W}} = \mathbf{D}^{-1/2} \mathbf{W} \mathbf{D}^{-1/2}, \quad (1)$$

with degree matrix $\mathbf{D}$, $\beta = 0.5$ and $s = 10^3$ (the logarithm compresses values that span four orders of magnitude). $c_i(P)$ is large for genes close to a perturbed gene in the graph and zero for genes more than three steps away.

2.3 GCP-Mamba

The model is summarised in Figure 1.

Tokens. Each of the G genes is one token. Token i is the sum of a learned identity vector, a projection of the gene’s control-cell

embedding $\mathbf{e}_i$ (its loadings on the top 64 principal components of control-cell expression, standardised) and of its control mean expression. A perturbation enters in three ways: (i) a learned flag vector added to the tokens of the targeted genes; all targets are measured genes, so no perturbation is lost (Section 3.1); (ii) a set embedding $\mathbf{z}_P = \sum_{g \in P} (f(\mathbf{e}_g) + \mathbf{t}_g)$ added to every token, where f is a two-layer MLP and $\mathbf{t}_g$ a per-target vector initialised at zero; only targets perturbed in training receive gradient for $\mathbf{t}_g$, so a target never seen in training is represented through $f(\mathbf{e}_g)$ alone, which is what permits zero-shot prediction; and (iii) the graph signal $c_i(P)$, linearly projected and added to token i .

Graph-conditioned selective scan. Each of two layers applies a forward and a backward Mamba-style selective SSM (Gu and Dao, 2023) with diagonal state matrix $\mathbf{A} = -\text{diag}(1, \dots, N)$, input-dependent $\mathbf{B}_i$ and $\mathbf{C}_i$, a SiLU output gate and a skip connection. The step size is conditioned on the graph,

$$\Delta_i = \text{softplus}(\mathbf{W}_\Delta \mathbf{u}_i + \mathbf{b}_\Delta + \mathbf{w}_c c_i(P)), \quad (2)$$

where $\mathbf{u}_i$ is the layer input and $\mathbf{w}_c \in \mathbb{R}^d$ is learned and initialised at zero, so that the model starts as an unconditional SSM. With zero-order-hold discretization, $\mathbf{h}_i = \exp(\Delta_i \mathbf{A}) \mathbf{h}_{i-1} + \Delta_i \mathbf{B}_i \mathbf{u}_i$; tokens near the perturbation take larger steps, writing more of their input into the state and forgetting more of the past. Because $\mathbf{w}_c c_i(P)$ differs between genes and between perturbations, permuting the graph changes the function the model computes.

Gene order. Genes are placed along the Fiedler vector (Fiedler, 1973) of the normalised Laplacian of $\mathbf{W}$, so that strongly co-expressed genes are close in the sequence.

Readout. The prediction for gene i is $\hat{\delta}_i = b_i + \mathbf{w}^\top \mathbf{h}_i + (\mathbf{U} \mathbf{h}_i)^\top (\mathbf{V} \mathbf{z}_P)$, where $\mathbf{h}_i$ is the final token state and $\mathbf{U}, \mathbf{V}$ have rank 16. The bilinear term expresses a gene-by-perturbation interaction and contains the low-rank linear model of Ahlmann-Eltze et al. (2025) as a special case. The bias b_i is initialised at the mean training response and $\mathbf{w}$ and $\mathbf{V}$ at zero, so that training starts from the *train-mean* baseline and learns deviations from it; without this initialisation, and without standardising $\mathbf{e}_g$, the model did not improve on the train-mean baseline in pilot runs on the validation set of split 1 (Supplementary Section S2).

Implementation and complexity. We implemented the scan in PyTorch as an exact chunked parallel scan: cumulative sums in log space within chunks of 16 tokens and a sequential recurrence across chunk states. Per-step log-decays are clamped at -5 (a decay factor of at least 0.0067) so that the within-chunk exponentials remain finite in single precision; the implementation matches a naive sequential recurrence to 10^{-5} (unit test in the repository). Time and memory are $\mathcal{O}(BGdN)$ for the scan and $\mathcal{O}(|\mathbf{W}|)$ for the sparse diffusion, i.e. linear in the number of genes; no $G \times G$ object is formed.

Training. $d = 32$, $N = 4$, dropout 0.1 in f ; AdamW (Loshchilov and Hutter, 2019) with learning rate 10^{-3} , weight decay 10^{-2} and batch size 8; gradient clipping at 1; mean-squared error over all genes. Models were trained for at least 40 and at most 100 epochs and stopped 15 epochs after the last improvement of the validation loss; the state with the lowest validation loss was kept. (The validation loss typically plateaus for several epochs before a model starts to use the perturbation; without the 40-epoch minimum, patience-based stopping truncated some runs during this plateau.) GCP-Mamba, the permuted-graph, graph-free, Graph-MLP and additive-prior variants were trained with three initialisations per split (seeds s , $s + 1000$, $s + 2000$ for split s ; for the permuted graph the permutation is redrawn with the seed); the other ablations with one. All runs used a 4-core CPU; a training run took up to 17 minutes (median 7).

2.4 Ablations and variants

All variants share the architecture, perturbation inputs (i)–(ii), optimiser and stopping rule, and differ in one factor. *Δ only*: the graph enters only through Δ , not through the tokens. *Δ only, $\mathbf{w}_c \sim \mathcal{N}(0, 1)$* : as before, but $\mathbf{w}_c$ is initialised randomly so that the conditioning is active from the start. *Permuted graph*: the gene labels of $\mathbf{W}$ are randomly permuted, which preserves its degree distribution and spectrum. *Random order*: genes are placed in random order instead of the Fiedler order. *Mamba (no graph)*: $c(P) \equiv 0$. *Graph-MLP*: the SSM layers are replaced by per-token MLPs, so genes interact only through $\mathbf{z}_P$ and the graph signal. *Additive prior*: the network predicts the residual $\delta - \pi$ with respect to the additive prior π (the sum of the targets’ observed single responses, or the mean single response for an unseen target; for single perturbations always the mean single response), with and without the graph.

2.5 Baselines

No change predicts $\delta = 0$. *Train mean* predicts the mean training response. *Additive* predicts the sum of the targets’ single responses observed during training (training or validation set), substituting the mean single response for a target without one. *Linear* is the model of Ahlmann-Eltze et al. (2025), $\mathbf{Y} \approx \mathbf{GMP}^\top + \mathbf{b}$, with gene embeddings $\mathbf{G}$ from the top 10 principal components of the training responses and each perturbation represented by the sum of its targets’ embeddings, taken either from $\mathbf{G}$ (*Linear (PCA)*, as in the original) or from the control-cell embeddings $\mathbf{e}_g$ that GCP-Mamba receives (*Linear (ctrl emb.)*); the ridge penalty was chosen on the validation set from $\{10^{-3}, \dots, 10^4\}$. *GEARS* was retrained with the official implementation and default hyperparameters (hidden size 64) on split 1 of the Norman data. Because GEARS trains on single cells and required about 40 minutes per epoch on our CPU, we trained it for 3 epochs instead of 20 (its learning rate halves after every epoch, so later epochs change the model little) and skipped its per-epoch evaluation of the whole training set and its built-in test-set evaluation, which are used only for logging and exceeded the available memory; its validation-based model selection was unchanged, and test predictions were made with `GEARS.predict` and scored like all others.

2.6 Metrics and statistics

Following Roohani et al. (2024), for every test perturbation we computed the mean squared error (MSE) and the Pearson correlation between predicted and observed δ on its top-20 differentially expressed (DE) genes. For double perturbations AB we computed the GI residual $\epsilon = \delta_{AB} - \delta_A - \delta_B$ from the measured singles (used only for scoring). The MSE of a predicted residual $\hat{\delta}_{AB} - \delta_A - \delta_B$ equals the MSE of $\hat{\delta}_{AB}$ itself, so it carries no extra information. We therefore report the fraction of GI variance explained, $R_{\text{GI}}^2 = 1 - \sum_{AB} \text{MSE}_{AB} / \sum_{AB} \epsilon_{AB}^2$, on “2/2 seen” doubles, where the additive baseline is exactly the measured-singles sum and scores 0, and the Pearson correlation between predicted and observed residuals used in earlier work. Metrics were averaged over initialisations, then over the test perturbations of a split and are reported as mean $\pm$ s.d. over the five splits. We compared GCP-Mamba with every other model using two-sided Wilcoxon signed-rank tests on per-perturbation differences pooled across splits, with Holm correction within each metric and test group (Supplementary Table S3). Because perturbations recur across splits these pooled tests are approximate, and we emphasise effect sizes and their consistency across splits.

3 Results

3.1 A perturbation encoding restricted to highly variable genes discards most perturbations

An earlier version of this study encoded a perturbation as a multi-hot vector over the 500 most highly variable genes (HVGs), the same genes whose expression was predicted, and parsed single perturbations from the condition name with a rule that mapped `ctrl+A` to the control condition. Only 12–17 of the 105 CRISPRa targets were among the HVGs (mean 14 over the ten splits used then), so 77–84% of training perturbations (mean 80%) and 70–94% of test perturbations received an all-zero input, and the 161–183 training perturbations of a split shared only 14–19 distinct input vectors (Supplementary Table S1; `gcpm/audit_hvg_encoding.py`). A model cannot distinguish perturbations it receives as identical inputs, so that version could at best learn the mean response; its conclusions about graph conditioning therefore said nothing about the architecture, in either direction. The problem is easy to miss: training converges, loss curves look normal, and the model is competitive with the train-mean baseline. It applies to any model whose perturbation input is indexed by the output gene set. Here every target is represented by its own token and by an embedding computed from control cells, and we recommend that perturbation models report how many targets reach the input.

3.2 GCP-Mamba does not outperform simple baselines

Table 1 and Figure 2 compare GCP-Mamba with the baselines and GEARS on the Norman data. As in previous benchmarks (Ahlmann-Eltze et al., 2025), the additive baseline was the most accurate predictor overall (MSE on the top-20 DE genes 0.224, averaged over all test perturbations), and for doubles with at least one seen target; for “2/2 seen” doubles its error (0.048) was a quarter of that of GCP-Mamba (0.196). GCP-Mamba (0.290 overall) was substantially more accurate than the train-mean (0.428), linear (0.349) and no-change (0.561) baselines.

GEARS could be retrained on only one split (Methods), so we compare it separately on split 1 (Table 5). There GEARS was the most accurate method for perturbations of unseen genes (unseen singles 0.214 vs. 0.262 for GCP-Mamba and 0.247 for the train mean; “0/2 seen” doubles 0.202 vs. 0.306), and it was more accurate than GCP-Mamba overall (0.230 vs. 0.269; Holm-adjusted $P = 0.003$). The additive baseline nevertheless remained the most accurate method overall on this split (0.213) and for doubles with seen singles (“2/2 seen”: 0.048 vs. 0.218 for GEARS). With one split and shortened training these GEARS results are indicative only, but they show that a graph-based model can predict unseen targets better than the mean response, which GCP-Mamba did not.

For unseen single perturbations, where no additive information exists, all methods except the no-change baseline and the linear model with control-cell embeddings were close together (MSE 0.260–0.281). GCP-Mamba (0.261) was not significantly more accurate than the train mean (0.270; Holm-adjusted $P = 0.42$), the linear model (0.269; $P = 1$), the graph-free Mamba (0.265; $P = 1$) or the model with a permuted graph (0.265; $P = 0.08$). In an earlier run with a single initialisation per split, GCP-Mamba had appeared significantly better than both graph controls on these perturbations; that difference was of the same size as the change caused by re-initialising GCP-Mamba alone (0.255 vs. 0.261), and it disappeared when the three initialisations were averaged (Supplementary Table S4). We report this because a single-seed comparison would have led us to claim a graph effect that is not there.

3.3 Neither the graph nor the step-size conditioning helps

Table 2 isolates the components of GCP-Mamba. Every variant without the graph in the token features was at least as accurate as GCP-Mamba

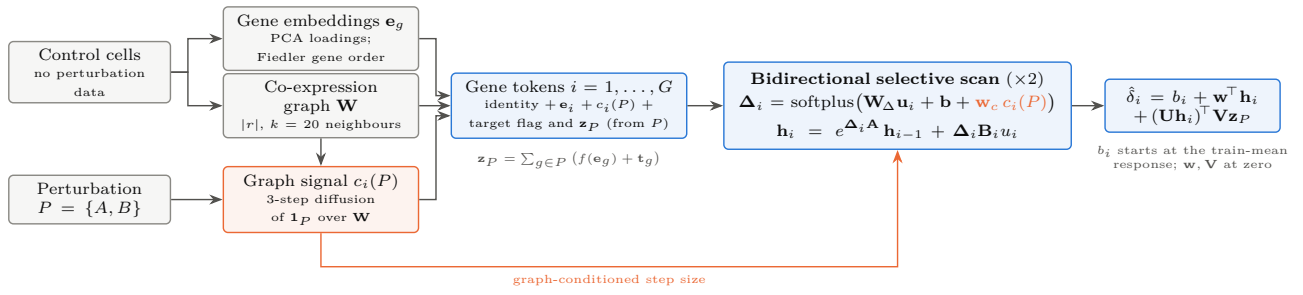
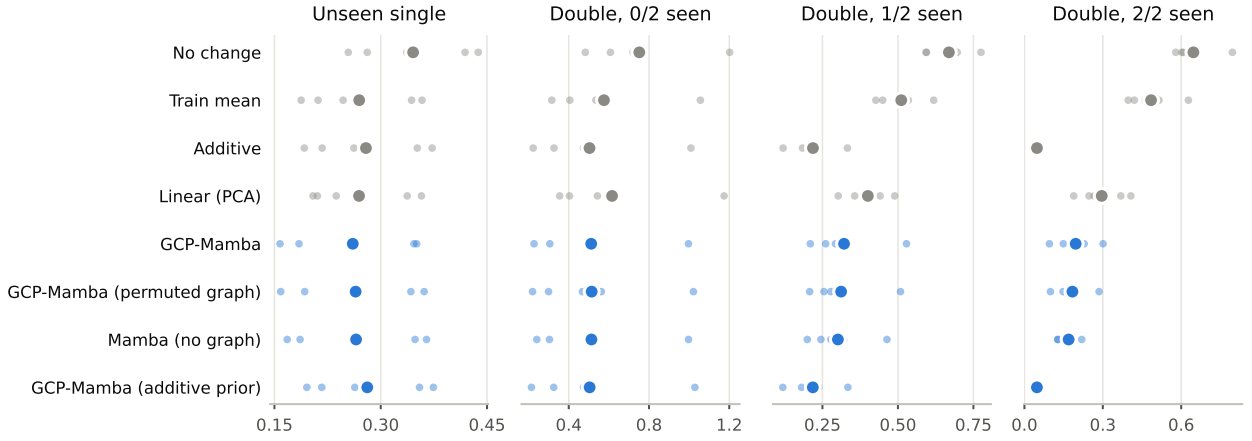


Fig. 1. GCP-Mamba. Everything derived from expression (grey) is computed from control cells only. The perturbation enters the gene tokens (a target flag and the set embedding $\mathbf{z}_P$) and, through its diffusion $c_i(P)$ over the co-expression graph (orange), the step size Δ_i of every token in the bidirectional selective scan. The bilinear readout starts at the train-mean prediction.

MSE, top-20 DE genes (lower is better)



Pearson r of δ , top-20 DE genes

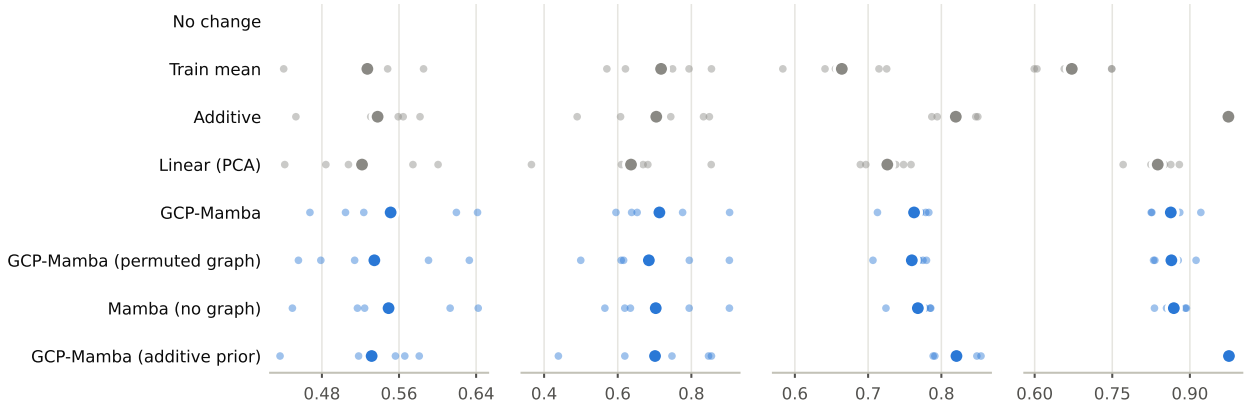


Fig. 2. Norman et al. CRISPRa data, five GEARS simulation splits. Each large dot is the mean over splits and each small dot one split (deep models averaged over initialisations); columns are test groups. Top: MSE on the top-20 DE genes of each test perturbation (lower is better). Bottom: Pearson correlation of predicted and observed change on the same genes (undefined for *No change*). Grey, baselines; blue, GCP-Mamba and variants. GEARS, run on one split only, is compared in Table 5.

overall: the graph-free Mamba (0.278), the Δ -only variant (0.279), the variant with randomly initialised conditioning weights (0.276) and the variant with a random gene order (0.278); with the permuted graph, the error was 0.285 (all differences from GCP-Mamba significant, Holm-adjusted $P \leq 1.1 \times 10^{-4}$; Supplementary Table S3). Conditioning the step size alone (Δ only) was indistinguishable from the graph-free model, although its conditioning weights moved away from their zero

initialisation ($\|\mathbf{w}_c\| = 0.48$ per SSM after training), and initialising the weights at random so that the step size depends strongly on graph proximity from the first update ($\|\mathbf{w}_c\| \approx 5.8$ after training) changed the error only marginally. Replacing the recurrence with per-gene MLPs did not change the error of the graph-conditioned model (0.288 vs. 0.290; $P = 0.83$), and the Fiedler ordering was not better than a random order.

Table 1. Norman et al. CRISPRa screen: MSE of the predicted expression change on the top-20 DE genes of each test perturbation (mean $\pm$ s.d. over the five GEARS simulation splits; lower is better; best in bold)

Model	Unseen single	Double, 0/2 seen	Double, 1/2 seen	Double, 2/2 seen	All test
No change	0.346 $\pm$ 0.081	0.752 $\pm$ 0.272	0.669 $\pm$ 0.077	0.647 $\pm$ 0.087	0.561 $\pm$ 0.061
Train mean	0.270 $\pm$ 0.077	0.576 $\pm$ 0.287	0.511 $\pm$ 0.076	0.485 $\pm$ 0.092	0.428 $\pm$ 0.061
Additive	0.279 $\pm$ 0.080	0.503 $\pm$ 0.303	0.219 $\pm$ 0.078	0.048 $\pm$ 0.006	0.224 $\pm$ 0.072
Linear (PCA)	0.269 $\pm$ 0.073	0.616 $\pm$ 0.328	0.400 $\pm$ 0.073	0.296 $\pm$ 0.090	0.349 $\pm$ 0.059
Linear (ctrl emb.)	0.322 $\pm$ 0.103	0.962 $\pm$ 0.788	0.518 $\pm$ 0.090	0.413 $\pm$ 0.122	0.452 $\pm$ 0.078
GCP-Mamba	0.261 $\pm$ 0.089	0.512 $\pm$ 0.300	0.321 $\pm$ 0.122	0.196 $\pm$ 0.078	0.290 $\pm$ 0.098

Table 2. Ablations and residual variants on the Norman data: MSE on the top-20 DE genes (mean $\pm$ s.d. over five splits; best in bold)

Model	Unseen single	Double, 0/2 seen	Double, 1/2 seen	Double, 2/2 seen	All test
GCP-Mamba	0.261 $\pm$ 0.089	0.512 $\pm$ 0.300	0.321 $\pm$ 0.122	0.196 $\pm$ 0.078	0.290 $\pm$ 0.098
GCP-Mamba (Δ only)	0.260 $\pm$ 0.085	0.490 $\pm$ 0.310	0.307 $\pm$ 0.112	0.173 $\pm$ 0.058	0.279 $\pm$ 0.093
GCP-Mamba (Δ only, $\mathbf{w}_c \sim \mathcal{N}(0, 1)$)	0.266 $\pm$ 0.091	0.492 $\pm$ 0.296	0.304 $\pm$ 0.088	0.154 $\pm$ 0.024	0.276 $\pm$ 0.081
GCP-Mamba (permuted graph)	0.265 $\pm$ 0.089	0.514 $\pm$ 0.314	0.312 $\pm$ 0.116	0.184 $\pm$ 0.069	0.285 $\pm$ 0.096
GCP-Mamba (random order)	0.260 $\pm$ 0.081	0.515 $\pm$ 0.253	0.304 $\pm$ 0.096	0.172 $\pm$ 0.067	0.278 $\pm$ 0.079
Mamba (no graph)	0.265 $\pm$ 0.090	0.513 $\pm$ 0.297	0.301 $\pm$ 0.101	0.169 $\pm$ 0.040	0.278 $\pm$ 0.088
Graph-MLP (no scan)	0.265 $\pm$ 0.090	0.515 $\pm$ 0.304	0.313 $\pm$ 0.098	0.196 $\pm$ 0.061	0.288 $\pm$ 0.085
GCP-Mamba (additive prior)	0.281 $\pm$ 0.080	0.504 $\pm$ 0.313	0.218 $\pm$ 0.079	0.048 $\pm$ 0.007	0.225 $\pm$ 0.073
Mamba (additive prior)	0.278 $\pm$ 0.087	0.506 $\pm$ 0.316	0.219 $\pm$ 0.079	0.048 $\pm$ 0.005	0.224 $\pm$ 0.075

Table 3. Prediction of genetic interactions (GI) in double perturbations whose two single perturbations were both observed during training (“2/2 seen”; top-20 DE genes; mean $\pm$ s.d. over five splits). R_{GI}^2 is the fraction of the variance of the GI residual ϵ that a prediction explains; an exactly additive prediction scores 0. The GI Pearson r (undefined for the additive baseline) ranks the models almost in reverse order of R_{GI}^2 .

Model	$R_{GI}^2 \uparrow$	GI Pearson $r \uparrow$
No change	-12.54 $\pm$ 1.47	0.25 $\pm$ 0.11
Train mean	-9.10 $\pm$ 1.26	0.28 $\pm$ 0.10
Additive	0.00 $\pm$ 0.00	—
Linear (PCA)	-5.14 $\pm$ 1.52	0.29 $\pm$ 0.12
Linear (ctrl emb.)	-7.47 $\pm$ 1.66	0.29 $\pm$ 0.10
GCP-Mamba	-3.04 $\pm$ 1.38	0.36 $\pm$ 0.08
Mamba (no graph)	-2.51 $\pm$ 0.60	0.37 $\pm$ 0.06
GCP-Mamba (additive prior)	0.01 $\pm$ 0.02	0.14 $\pm$ 0.04
Mamba (additive prior)	-0.01 $\pm$ 0.06	0.20 $\pm$ 0.07

Table 4. Adamson et al. CRISPRi screen, unseen single perturbations: MSE and Pearson r of the predicted change on the top-20 DE genes (mean $\pm$ s.d. over five GEARS simulation splits).

Model	MSE $\downarrow$	Pearson $r \uparrow$
No change	0.361 $\pm$ 0.051	—
Train mean	0.171 $\pm$ 0.065	0.79 $\pm$ 0.10
Additive	0.170 $\pm$ 0.064	0.79 $\pm$ 0.10
Linear (PCA)	0.161 $\pm$ 0.051	0.81 $\pm$ 0.06
Linear (ctrl emb.)	0.167 $\pm$ 0.065	0.79 $\pm$ 0.10
GCP-Mamba	0.166 $\pm$ 0.064	0.79 $\pm$ 0.10
GCP-Mamba (Δ only)	0.164 $\pm$ 0.060	0.79 $\pm$ 0.10
GCP-Mamba (Δ only, $\mathbf{w}_c \sim \mathcal{N}(0, 1)$)	0.164 $\pm$ 0.060	0.79 $\pm$ 0.10
GCP-Mamba (permuted graph)	0.165 $\pm$ 0.062	0.79 $\pm$ 0.10
GCP-Mamba (random order)	0.162 $\pm$ 0.063	0.78 $\pm$ 0.10
Mamba (no graph)	0.162 $\pm$ 0.062	0.79 $\pm$ 0.10
Graph-MLP (no scan)	0.162 $\pm$ 0.062	0.80 $\pm$ 0.10
GCP-Mamba (additive prior)	0.166 $\pm$ 0.064	0.79 $\pm$ 0.10
Mamba (additive prior)	0.162 $\pm$ 0.062	0.79 $\pm$ 0.10

In short, none of the components that distinguish GCP-Mamba from a generic model contributed measurably to accuracy.

3.4 No model predicts genetic interactions beyond additivity

Double perturbations are interesting chiefly for their non-additive component. When the network predicted only the deviation from the additive prior, it matched the additive baseline (0.225 vs. 0.224 overall; Table 2) and, on “2/2 seen” doubles, explained none of the GI variance ($R_{GI}^2 = 0.005 \pm 0.025$ for GCP-Mamba with the prior, -0.005 ± 0.061 without the graph; Table 3): it learned a negligible correction to the additive prediction. All deep models without the prior had strongly negative R_{GI}^2 (from -2.2 to -3.1), i.e. their errors on these doubles were two to three times larger than the interaction itself.

The GI Pearson correlation, used to assess GI prediction in earlier work including the first version of this study, ranked the models almost in reverse (Fig. 3). It was highest (0.36–0.39) for the models whose R_{GI}^2 was most negative, lowest for the additive-prior models (0.14 and 0.20), and gave the trivial no-change and train-mean predictors scores (0.25 and 0.29) comparable to the linear model. Predicting no change yields a predicted residual of $-(\delta_A + \delta_B)$, which is correlated with the true residual whenever interactions are buffering (sub-additive); in general, any prediction that is shrunk towards zero inherits this correlation, whereas a prediction close to the additive sum has a small, noisy residual whose correlation is low. A correlation between residuals therefore should not be used alone to claim GI prediction; a variance-explained measure relative to the additive prediction avoids the problem.

Table 5. Norman data, split 1 only, including GEARS (retrained with the official implementation for 3 epochs; Methods): MSE on the top-20 DE genes (best in bold).

Model	Unseen single	Double, 0/2 seen	Double, 1/2 seen	Double, 2/2 seen	All test
No change	0.337	0.608	0.696	0.647	0.559
Train mean	0.247	0.405	0.525	0.515	0.420
Additive	0.262	0.326	0.220	0.048	0.213
Linear (PCA)	0.237	0.403	0.441	0.369	0.356
Linear (ctrl emb.)	0.247	0.390	0.491	0.468	0.396
GEARS	0.214	0.202	0.254	0.218	0.230
GCP-Mamba	0.262	0.306	0.293	0.207	0.269
Mamba (no graph)	0.259	0.304	0.278	0.186	0.257
GCP-Mamba (additive prior)	0.263	0.325	0.219	0.048	0.213

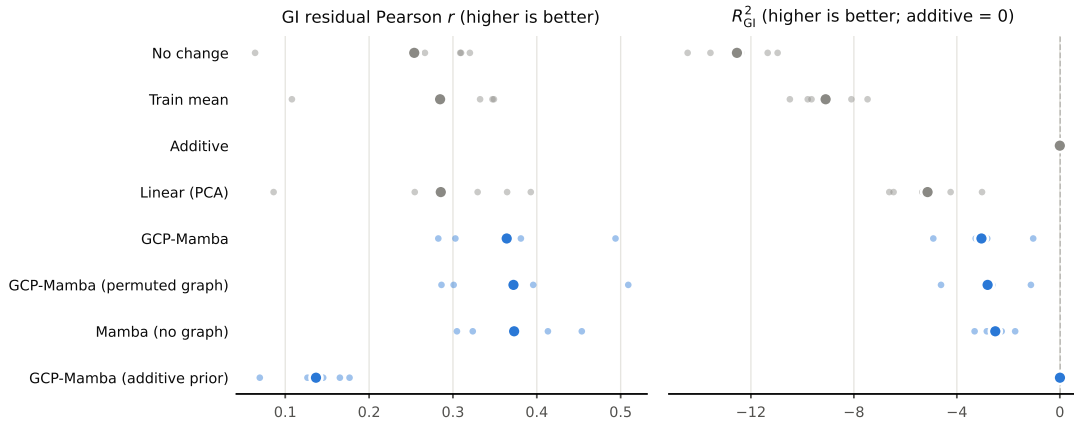


Fig. 3. Genetic-interaction prediction on “2/2 seen” doubles of the Norman data. Left: Pearson correlation between predicted and observed GI residuals (undefined for the additive baseline, whose predicted residual is zero). Right: fraction of GI variance explained, R^2_{GI} (the additive baseline is 0 by construction). Dots as in Fig. 2.

3.5 Adamson CRISPRi screen

On the Adamson data, which contain only single perturbations, all methods other than the no-change baseline had MSEs between 0.161 and 0.171 and Pearson correlations between 0.78 and 0.81, differences far smaller than the variation between splits (s.d. 0.05–0.07; Table 4). No method was distinguishable from the train mean (0.171), in line with the low perturbation-specific variance of this dataset reported by Csentesi et al. (2025).

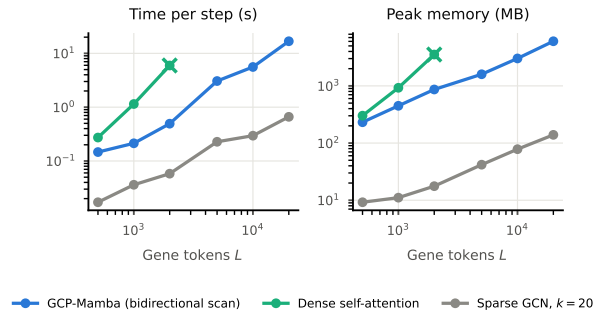


Fig. 4. Measured cost of one training step versus the number of gene tokens L (log–log). Crosses mark the last size at which dense self-attention fitted in 15 GB of memory.

3.6 Computational cost

Figure 4 shows the measured time and peak memory of one training step (two layers, batch size 8, forward and backward pass, 4-core CPU) as a function of the number of gene tokens. GCP-Mamba grew approximately linearly, from 0.15 s and 0.23 GB at 500 genes to 3.1 s and 1.6 GB at 5,000 genes (the size of the Norman gene panel) and 16.8 s and 6.0 GB at 20,000 genes. Dense self-attention needed 3.5 GB at 2,000 genes and ran out of the 15 GB available from 5,000 genes. A two-layer sparse graph convolution on a 20-nearest-neighbour graph, the kind of computation used by GEARS, was, however, 6–25 times faster and used 25–50 times less memory than the scan (the ratio grows with size for time and stays between 25 and 49 for memory). Linear-time SSMs therefore make transcriptome-wide token models feasible where attention is not, but they offer no efficiency advantage over sparse graph networks.

4 Discussion

We set out to test whether conditioning a linear-time state-space model on a gene–gene graph yields better predictions of genetic perturbation responses than simple baselines. With inputs that represent every perturbation, a training procedure that starts from the train-mean prediction, controls that isolate each component and three initialisations per split, the answer is no. For double perturbations, the sum of single effects remained the best predictor, consistent with Ahlmann-Eltze et al. (2025), and none of GCP-Mamba’s variants explained any of the variance

of genetic interactions beyond it (nor did GEARS on the one split where we could run it). For unseen single perturbations, GCP-Mamba and its variants stayed within a few per cent of the mean response, whereas GEARS improved on it. The mechanism that motivated the architecture—modulating the SSM step size by graph proximity to the perturbation—had no measurable effect, and removing the graph altogether did not make predictions worse.

We see two reasons why the graph did not help. First, the explainable signal in these screens is low-dimensional: the Norman responses are dominated by a few differentiation programmes in K562 (Norman et al., 2019), and the Adamson responses by the unfolded protein response (Adamson et al., 2016), so a low-rank readout of a perturbation embedding captures most of what can be predicted, leaving little for a gene-specific propagation mechanism to add. Second, co-expression in unperturbed cells may say little about how a specific perturbation propagates. GEARS, which represents perturbations through a Gene Ontology graph, was the only method that predicted unseen targets clearly better than the mean response, which suggests that functional-annotation or regulatory graphs (Roohani et al., 2024; Wenkel et al., 2026) carry information that control-cell co-expression does not; our code allows any graph to be substituted.

Our experience carries four practical lessons for developing and evaluating such models. (i) Audit the input encoding: an encoding tied to the output gene set gave most perturbations an all-zero input in an earlier version of this work, without any visible symptom in training. (ii) Initialise deep models at, and always compare them with, the train-mean and additive predictors; without that initialisation our model did not improve on the train mean at all. (iii) Use several initialisations: a single-seed comparison suggested a graph effect that did not survive replication. (iv) Measure genetic-interaction prediction as variance explained relative to the additive prediction on doubles whose singles were observed; correlations between residuals can rank models in reverse.

Limitations. We evaluated two K562 screens with the standard GEARS splits; screens in other cell types, genome-wide single-perturbation screens and larger combinatorial libraries may behave differently, and the Replogle et al. screens could not be included because most of their targets are not among the measured genes in the GEARS release, which our token-based encoding requires. Hyperparameters were chosen on the validation set of the first split only and not tuned per variant; a larger search could change absolute errors, although the additive baseline was not approached by any variant. We used a single co-expression graph computed from control cells. Because of its cost on CPU, GEARS was retrained on only one split and for 3 instead of 20 epochs, so its numbers are indicative rather than a full benchmark. Our PyTorch scan runs on CPU; a fused GPU kernel would change absolute run times but not the linear scaling.

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Author contributions

A.P. designed the model and performed the computational experiments; R.H. contributed to the biological interpretation and to writing. Both authors approved the final manuscript.

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Data availability

The Norman et al. and Adamson et al. data are available from NCBI GEO (GSE133344 and GSE90546) and in processed form from the GEARS repository. Code, split definitions, per-perturbation metrics and scripts that regenerate every table and figure are available at <https://github.com/The-MathKing/gcpmamba>; predictions of all models will be deposited in Zenodo.

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